DESIGN OF VERTICAL AXIS WIND TURBINE BLADES AND MOUNT

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I. PROBLEM STATEMENT

A wind turbine is a device that generates power using blades spun by wind[2]. There are two main types of wind turbines: a horizontal axis turbine and a vertical axis turbine[2]. Our group set out to design a vertical axis wind turbine that has a sweep area of at most 80 centimeters^2. In addition, the motor must be concealed inside a mount, and the blades must be detachable from a blade mount. The blades and mounts will be 3D printed using a Prusa 3D Printer. The cost of the printed materials will be determined by how much material is used and how long it takes to print. The rest of the materials used have been provided, and their costs will be considered as well. When we have finished our design, we will be comparing the efficiency of our design to the efficiency of a pre-made design. This will be done by using a hairdryer as the source of wind, measuring the voltage and current of both designs using a multimeter.

II. CONCEPT

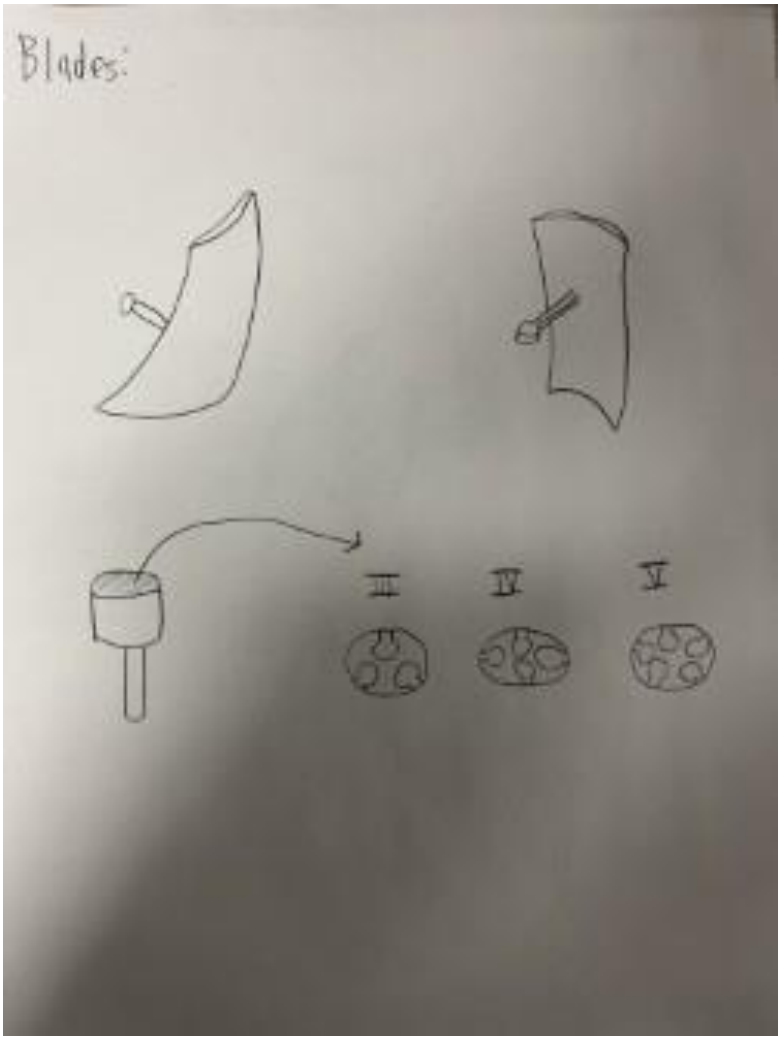


Figure 1: Our initial concept ideas for the mount

There are many different types of blades that can be used for a vertical axis wind turbine. We chose to use a helical shape for the blades because we thought it looked cool. We designed the blades in SolidWorks with some guidance [1]. Designing the blade also included deciding how many blades we wanted. We went with five blades because we found it to have a better performance [2].

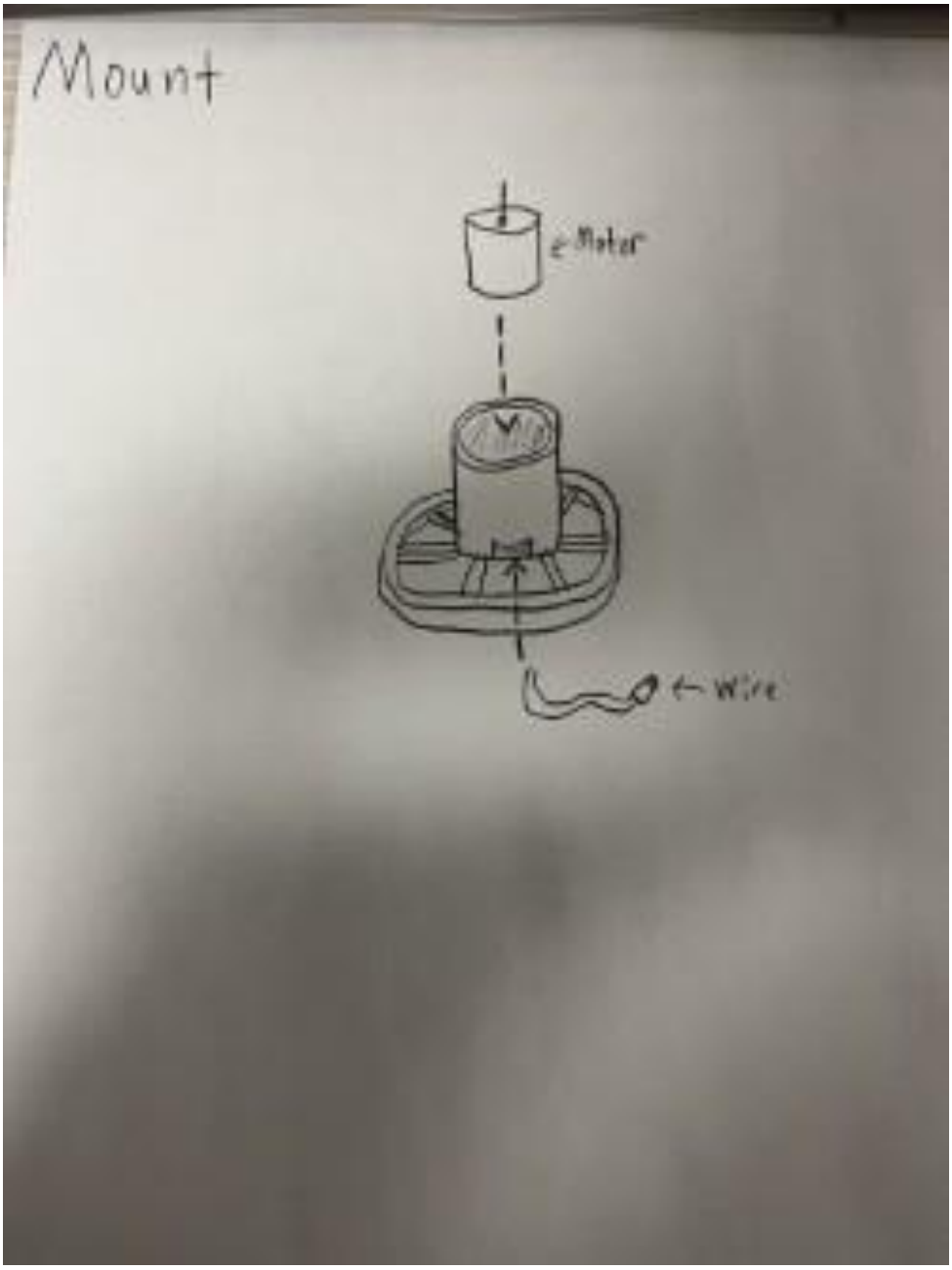


Figure 2: Our initial concept ideas for the mount

The motor mount was designed to be enclosed entirely with a hole on the side to allow for the wires to exit the mount. We later modified the initial concept to have ribs for further support and a stronger base.

III. DESIGN

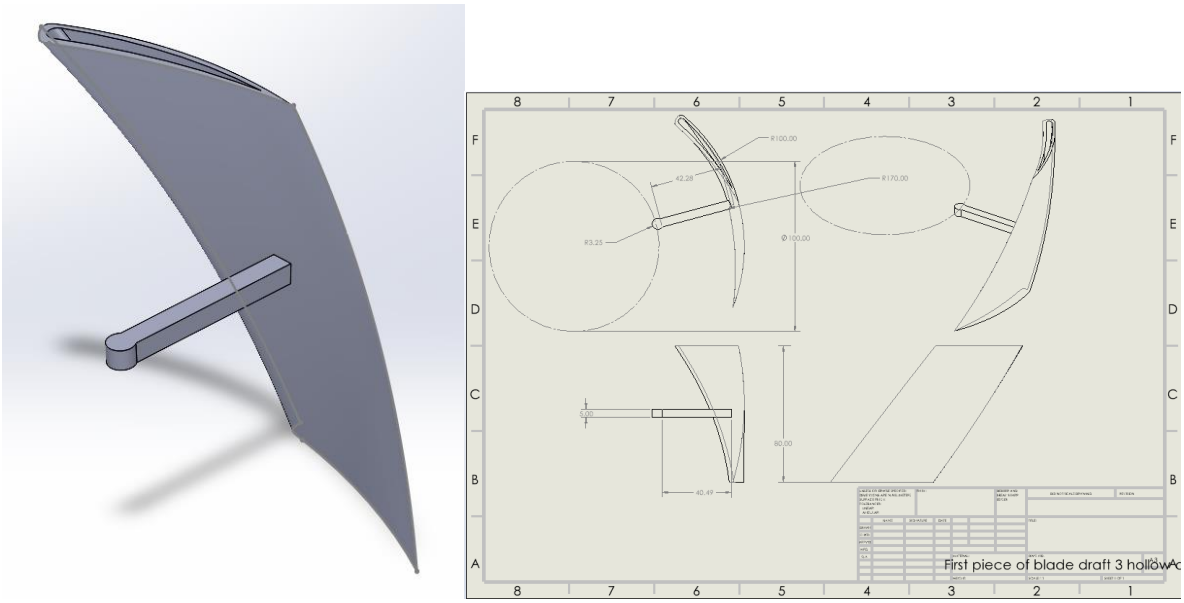


Figure 3-4: Model and drawing of the blades

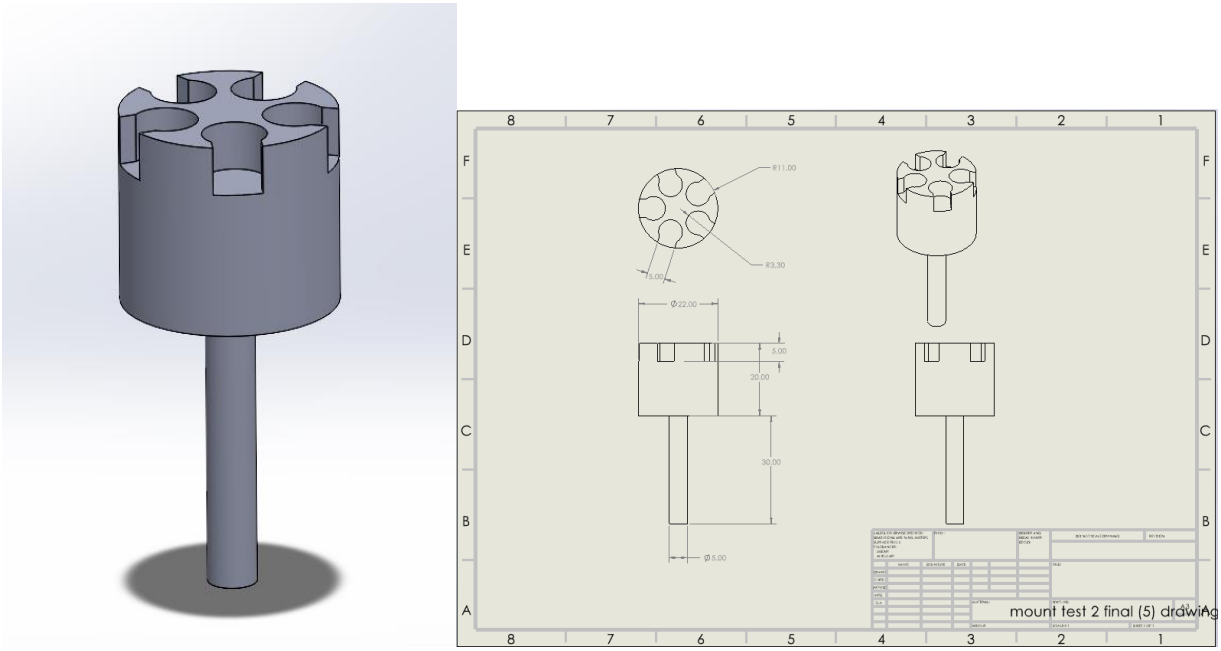


Figure 5-6: Model & drawing of the blade mount

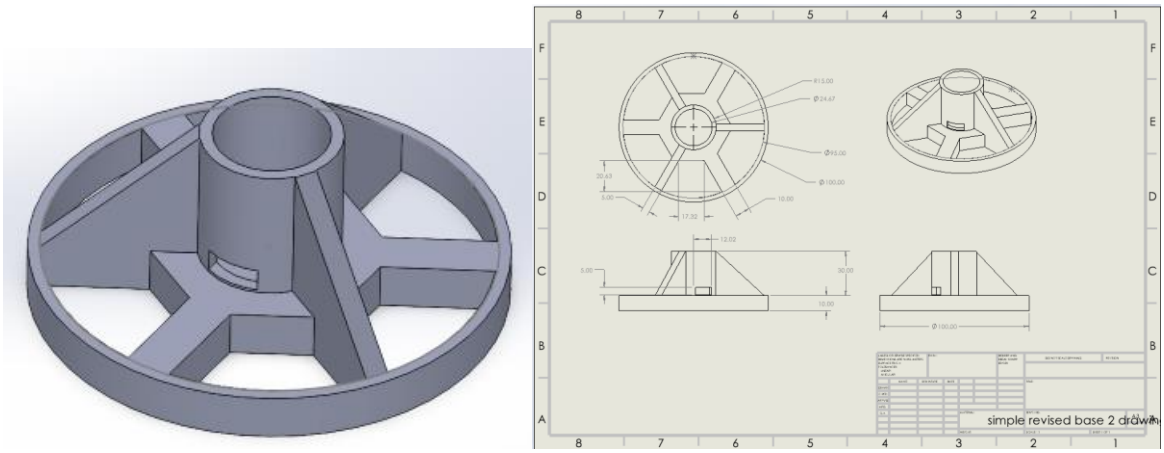


Figure 7-8: Model and drawing of the motor mount

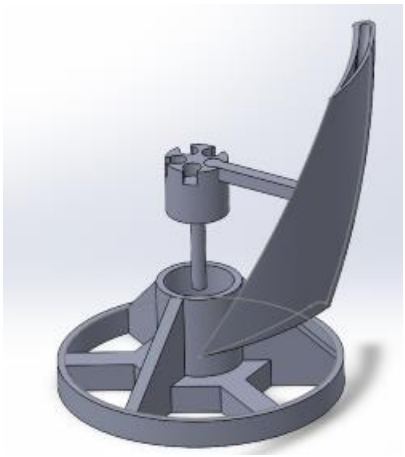


Figure 9: Model of the final assembly

$$r = 42.28\,mm + 3.25\,mm = 45.53\,mm$$

$$D = 2r = 2(45.53mm) = 91.06\,mm = 9.106\,cm$$

$$H = 80\,mm = 8\,cm$$

$$SWEEP\ AREA = DH = (9.106cm)(8cm) = 72.848\,cm^2$$

Figure 10: Equation used to determine sweep area of the turbine

IV. BILL OF MATERIALS

Wind Turbine Material List			
Number	Item Name	Quantity	Source
1	DC 0.1V-5.5V 100-6000RPM Motor	1	Provided By Dr. Alam
2	Motor Mount	1	3D Printed Using Prusa 3D Printer
3	Blade Mount	1	3D Printed Using Prusa 3D Printer
4	Turbine Blades	5	3D Printed Using Prusa 3D Printer
6	Test Blade Mount	1	Provided By Dr. Alam
7	Test Blades	4	Provided By Dr. Alam

Figure 11: List of materials used in our design with available sources

V. COST ANALYSIS

Vertical wind turbines that have spiral blades like what we 3D printed range from $80 to $550. Our 3D printed wind turbine only costs $31.13 which is at least $48.87 cheaper. However, the printing time for our wind turbine was a little more than 5 hours per print (3 items) which resulted in about a 15-hour print in total. This is because we had hollowed our blades and the blades are very precise. However, the amount that is saved from 3D printing is enough to reason the time.

Wind Turbine Cost Analysis

Basic Materials

Item	Cost
DC Motor	$12.99
Printing Filament	$20 / kg
Printing Time	$1 / hr

3D Prints

Print #	Time (hr)	Material (kg)	Cost ($)	
1	5.28	0.06	6.48	> 2 Wings & Blade Mount
2	5	0.055	6.1	> Motor Mount & 1 Wing
3	4.68	0.044	5.56	> 2 Wings

TOTAL 31.13

Figure 12: A breakdown of the costs of our design

VI. RESULTS

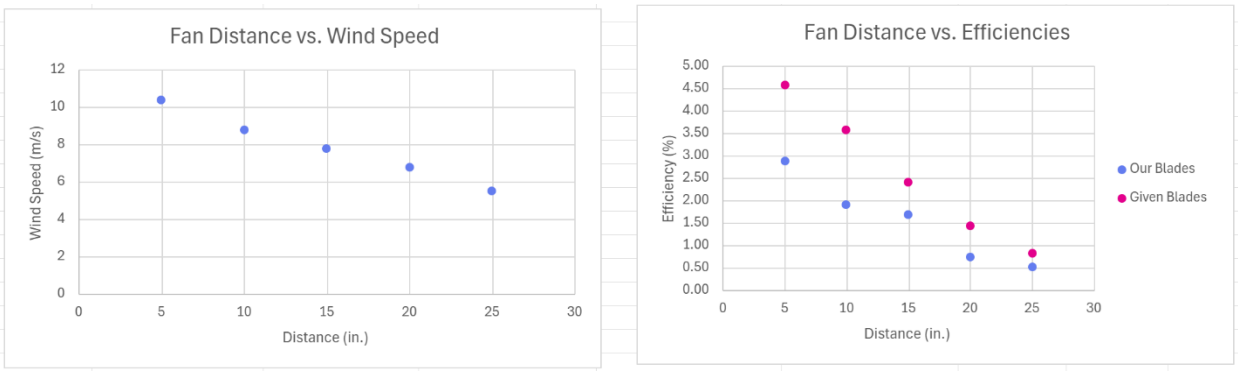


Figure 13: Test of the wind turbines, displaying the wind speeds at a certain distance and the efficiencies of the turbines at those distances



Figure 14: Our final vertical axis wind turbine

Our final design reached a maximum voltage of 6 V and a maximum current of 26.5 mA. Compared to the blades given, our design was able to achieve a higher voltage but not a higher current. The power output of the final design, compared to the power generated by the wind, gave a max efficiency of 2.88%. When testing our design, we found that while it took some time to reach its maximum value, the values would remain consistent for the most part. Compared to the given blades, the values tended to fluctuate more.

VII. DISCUSSIONS

This final project was easier and more difficult than our midterm project. It was easier because we had an idea early on and were able to be way more relaxed throughout the whole experience. However, we decided to make a spiral style blade for our vertical wind turbine. This took a lot more time than expected, however it was worth the added time.

The biggest thing that was noticed during this project is that our wind turbine was better than we expected at times. When we measured at 5 inches away the turbine created enough voltage that the multimeter could not record the output. While it created more voltage than we thought as the wind turbine was moving further away from the hairdryer, we had a harder time getting the turbine to work fully. Overall, our wind turbine created a good amount of voltage, but our wind turbine needed a larger wind source.

ACKNOWLEDGEMENTS

1. SolidWorks 3D Software, which aided in the design of the blades and mounts.
2. Prusa 3D Printer & Software, which aided in the construction of the blades and mounts.
3. Dr. Md Erfanual Alam, who provided the resources needed for the designs.
4. Mr. John Gidlund, who helped with tolerance analysis of the motor mount.

REFERENCES

[1]3D SolidWorks. (2024, August 20). How to make a vertical wind turbine in SolidWorks. YouTube. https://www.youtube.com/watch?v=0wlc4kG7YWY

[2]Suastawa, I. W., Santika, I. W. G., & Rusmariadi, I. K. (2024). Effect of Number of Blades on Performance in Vertical Axis Wind Turbine Three-Phase Permanent Magnet Model. Advances in Engineering Research, 570–579. https://doi.org/10.2991/978-94-6463-587-4_64